

5.2

Exercițiul 2 L N vectorilor lui "găsim" $x-y-z=0$ a la marea f_1

a $v_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ L marea găsim cînd marea

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Avem se A marea la marea 3 v_1 a la marea se găsim del A

del A = $1+0+1-0-0-1=1 \neq 0$ marea L N a marea la marea

Pentru v_1, v_2 marea marea v_1 a v_2 marea marea v_3 a marea marea
 $f_A(v_1) = 2v_1, f_A(v_2) = 2v_2, f_A(v_3) = 3v_3$

Transformarea marea L marea marea

$[v_1]_B = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, [v_2]_B = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, [v_3]_B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ Pentru marea marea marea marea

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Pentru marea A marea marea marea marea marea.

$$C = ABA^{-1}$$

A marea marea a A^{-1} marea marea

$$A^{-1} = \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right) = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix}$$

Exercice 1

$$C = AB^{-1} \quad \left(\begin{array}{ccc|ccc} 0 & 1 & -1 & 0 & 1 & -1 \\ -1 & 1 & 0 & 0 & 1 & 1 \\ 1 & -1 & 1 & 0 & 0 & 0 \end{array} \right)$$

a) $BA^{-1} = \left(\begin{array}{ccc|ccc} 2 & 0 & 0 & 0 & 2 & -2 \\ 0 & 2 & 0 & -2 & 2 & 0 \\ 0 & 0 & 3 & 3 & -3 & 3 \end{array} \right)$

c) $A \cdot (BA^{-1}) =$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 2 & -2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 0 & 1 & 1 & 3 & -3 & 3 \end{array} \right) \quad C = \left(\begin{array}{ccc|ccc} 3 & -1 & 1 & 0 & 2 & -2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 1 & -1 & 3 & 3 & -3 & 3 \end{array} \right)$$

Donc les valeurs propres

$\vec{v}_1 \in C(1,1,1)^T = \{3-1+1, 1+1+1, 1-1+3\}^T = \{3, 3, 3\}^T = 3(1,1,1)^T$

$\vec{v}_2 \in C(0,1,1)^T = (0-1+1, 0+1+1, 0-1+3)^T = 2(0,1,1)^T$

$\vec{v}_3 \in C(1,1,0)^T = (3-1+0, 1+1+0, 1-1+0)^T = 2(1,1,0)^T$

$\vec{v}_1, \vec{v}_2, \vec{v}_3$ forment une base de $\mathbb{R}^3$ car ils sont linéairement indépendants. On a donc une base de $\mathbb{R}^3$ formée de $\vec{v}_1, \vec{v}_2, \vec{v}_3$.